

Indian Statistical Institute
Second Semester Examination 2003-2004
B.Math (Hons.) I Year
Discrete Mathematics and Number Theory
Time: 3 hrs Date:10-05-04 Max. Marks : 100

Remark: Each question carries 20 marks. Answer any five questions.

1. Let p be an odd prime. Let B be a set of integers such that (i) p does not divide any element of B and (ii) for every integer m such that p does not divide m , there is a unique $b \in B$ such that $m \equiv \pm b \pmod{p}$. Such a set B is called a half system modulo p .
 - (a) Give an example of a half system modulo p .
 - (b) For any integer m such that p does not divide m , show that there is a unique permutation $\pi_m : B \rightarrow B$ and a unique function $e_m : B \rightarrow \{+1, -1\}$ such that $mb \equiv \pi_m(b)e_m(b) \pmod{p}$ for every $b \in B$.
 - (c) Show that the number $\prod_{b \in B} e_m(b)$ depends only on p and m and not on the half system B .
2. (a) Let $\{a_n : n = 0, 1, 2, \dots\}$ be the sequence defined by $a_0 = 1, a_1 = 2, a_{n+1} = \frac{a_n^2 + 1}{a_{n-1}}$ for $n \geq 1$. Show that all terms of this sequence are integers.
 - (b) Let x, y be positive integers such that $x|y^2 + 1$ and $y|x^2 + 1$. show that either $x = y = 1$ or x, y are consecutive terms of the sequence in part (a).
(Hint: Use induction for both parts. For the second part show that if $x < y$ then $n := \frac{x^2 + y^2 + 1}{xy}$ and $x_0 := ny - x$ are both positive integers. Further show that x_0 & y also satisfy the hypothesis)
3. Let $\mathcal{R}$ be the ring of all arithmetic functions with point-wise addition and convolution. If $f \in \mathcal{R}$ is a multiplicative function (not identically zero) then show that the multiplicative inverse of f exists and is multiplicative.
4. Let $\mathcal{X}$ be an incidence system with exactly 24 points such that any 5 distinct points of $\mathcal{X}$ are together incident with a unique block of $\mathcal{X}$ and each block of $\mathcal{X}$ is incident with exactly 8 points. Find the total number of blocks in $\mathcal{X}$. Show that any two blocks of $\mathcal{X}$ are together incident with an even number of points.
5. Let $\mathcal{X}$ be an incidence system such that any two distinct points of $\mathcal{X}$ are together incident with exactly λ blocks (where $\lambda > 0$ is a constant) and no point is incident with all the blocks. Then show that $\mathcal{X}$ has at least as many blocks as points.
6. Prove that every element of a finite field can be written as a sum of two squares. Show that if $q \equiv 3 \pmod{4}$ then the translates of the set of non-zero squares in $\mathbb{F}_q$ form the blocks of a 2-design. Compute its parameters.